

YouTube Trending Videos Analysis

Business Insights & Executive Report

Project Objective

Objective: The purpose of this project was to understand what drives trending content on YouTube. Using **SQL**, **Python**, and **Power BI**, the analysis focused on audience reach, engagement, channel dominance, category performance, and viral content behavior.

Business Problem

Millions of videos are uploaded to YouTube, yet only a small percentage reach the trending page. The challenge was to identify whether success is driven by **views**, **engagement**, **channel reputation**, or **content category**.

Key Insight 1: Channel Dominance

The analysis revealed that channels such as **T-Series**, **Marvel Entertainment**, and **FoxStarHindi** captured a significant share of total views. This indicates that audience attention is highly concentrated among a small group of creators.

Key Insight 2: Entertainment Drives Reach

Category-level analysis showed that **Entertainment** generated the largest share of views. This demonstrates that entertainment content remains one of the strongest drivers of audience acquisition.

Key Insight 3: Views and Likes Move Together

The correlation analysis showed a strong positive relationship between **Views** and **Likes**. As videos reached larger audiences, audience appreciation generally increased as well.

Key Insight 4: Reach Does Not Always Mean Engagement

Several videos crossed the **1 Million View** mark but recorded relatively low engagement rates. This proves that **reach alone should not be used as the primary success metric**.

Key Insight 5: Engagement Reflects Content Quality

Videos with higher engagement rates generated stronger audience interaction through likes and comments. Engagement provides a more complete understanding of audience interest than views alone.

Business Recommendations

- 1 **Track Engagement Alongside Views** – Evaluate success using both reach and engagement metrics.
- 2 **Partner With Consistently Trending Channels** – Channels with repeated appearances demonstrate audience loyalty.
- 3 **Use Category Insights for Content Planning** – Focus on categories aligned with business goals.
- 4 **Monitor Engagement Rate** – Engagement helps identify content quality and audience connection.
- 5 **Focus on Long-Term Audience Interaction** – Encourage likes, comments, and participation.

Conclusion

This project demonstrates an **end-to-end Data Analytics workflow** involving data cleaning, SQL-based business analysis, Python exploratory analysis, Power BI dashboard development, and business storytelling. The key learning from this analysis is that **views measure visibility, while engagement measures audience connection**. Organizations that focus on both metrics are more likely to build sustainable content strategies.